

UNIVERSIDAD POLITÉCNICA DE AGUASCALIENTES

BACHELOR'S DEGREE IN ENGINEERING TECHNOLOGIES IN
INFORMATION TECHNOLOGY AND DIGITAL INNOVATION



Sprint #4 Deliverable 5

KAVEM System - Inventory & Production Module

COURSE

Project II

Period

2026 - 2

TEAM #3

PROFESSOR

Humberto Cruz Aguilar

TEAM MEMBERS

Olivia Lizbeth Chairez Alba	UP240594
Noé Ezequiel Estrada Ramírez	UP240770
Omar Facio Palos	UP240470
Alan Ricardo Jauregui De La Riva	UP240960

GROUP

TIID05B

Aguascalientes, México

June 07th, 2026

Contents

1	Introduction	2
2	Functional and Non-Functional Requirements	3
2.1	Functional Requirements (FR)	3
2.2	Non-Functional Requirements (NFR)	4
2.3	Color Palette	5
3	Use Case Diagram	7
4	Use Case Scenario Cards	8
4.1	Card 1: Enter product/stock	8
4.2	Card 2: Edit product information	9
4.3	Card 3: Check availability	9
5	Database Entity-Relationship Diagram	10
5.1	Main Entities	10
6	Interface Design (Mockups)	11
6.1	Main Inventory Dashboard Mockup	11
6.2	Type of products Mockup	11
6.3	Product Details Mockup	12
6.4	Raw materials Mockup	12
7	Navigation Design	14
7.1	Navigation Structure	15
8	Test Cases	16
8.1	Test Cases for Inventory Management	16
8.2	Test Summary for Sprint #4	18
9	Screenshots of the Program in Operation	19
9.1	Screenshot 1: Product Main	19
9.2	Screenshot 2: Products Part	19
9.3	Screenshot 3: Materials	20
10	Sprint Management Using Jira	21
10.1	Sprint #3 Overview	21
10.2	Jira Board Screenshot	21
10.3	Tasks Completed in Sprint #4	22
10.4	Task Assignments Summary	22
10.5	Burndown Chart	22

Introduction

The following content presents the results and advances of Sprint #4. In this sprint, we transitioned into the development and implementation of the Inventory & Production module. We defined and structured the functional requirements (FR01-FR14, FR31-FR33) concerning catalogs, attributes, pricing, logs, and synchronization. Furthermore, we designed the use case diagrams, scenario cards, database entity-relationship models, and system interfaces. Finally, we executed comprehensive test cases tailored to resource handling and analyzed performance workflows within Jira. This document consolidates all deliverables for the current sprint.

Functional and Non-Functional Requirements

Functional Requirements (FR)

Product Code & Name (CRUD Catalog)

- **FR01** - The system must record a unique identification code for each product.
- **FR02** - The product code must have an exact length of 6 alphanumeric characters.
- **FR03** - The system must record the product name.
- **FR04** - The product name must be unique.
- **FR05** - The product name must have a maximum length of 40 characters and a minimum length of 3 characters.
- **FR06** - The product name is a mandatory field (Must not be left blank).

Product Attributes (Size & Category)

- **FR07** - The system must obligatorily break down each product by category type: Life Jacket, Swimming Board, or Slider.
- **FR08** - If the product is a Life Jacket, the system must require the selection of a valid size from the following options: Flea, Infant, Junior, Medium, Adult, Extra Large.
- **FR09** - If the product is a Swimming Board, the system must automatically assign the property “One Size”.
- **FR10** - If the product is a Slider, the system must require the selection of a valid size from the following options: Small, Junior, Medium, Adult.

Costs & Base Prices

- **FR11** - The system must record the official base price of each product (VAT and freight not included).
- **FR12** - The product base price must accept positive numeric decimal values only.
- **FR13** - The base price is a mandatory field and cannot be equal to zero.
- **FR14** - The system must record the number of available units in inventory.

Logistics, History & Search

- **FR31** - The system must recalculate and update the projected delivery times to the customer by directly linking current manufacturing times with the real availability of supplies in inventory.

- **FR32** - The system must record a historical log of all entries, outputs, and manual adjustments made to the inventory, including user, date, and reason.
- **FR33** - The system must have advanced search filters in the inventory dashboard to locate products by name, category, size, or production status.

Non-Functional Requirements (NFR)

- **NFR01** - The interface dedicated to the inventory and production module must visually incorporate the KAVEM company logo and the established institutional color palette.
- **NFR02** - To achieve a more agile and familiar workflow for warehouse and production employees, the system must employ recognizable, standardized icons on navigation links and action buttons (e.g., add, edit, delete, or alert buttons) instead of extensive text labels.
- **NFR03** - The colors used in the inventory screens must exactly match the visual identity; the hexadecimal codes selected for the program are: #E49C54, #E6BE8C, #F2B06E, #F5ECE3.
- **NFR04** - The processing and update time of any physical change in inventory data must be reflected across sales and administration screens within a maximum of 15 minutes (ideally in real time to mitigate double-entry errors).

Color Palette



Figure 1: KAVEM Color Palette: #E49C54, #E6BE8C, #F2B06E, #F5ECE3

Use Case Diagram

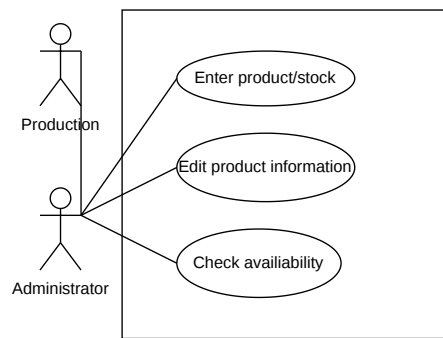


Figure 2: Use Case Diagram for Inventory Module

Main Actors:

- Administrator
- Production

Main Use Cases:

- Enter product/stock
- Edit product information
- Check availability

Use Case Scenario Cards**Card 1: Enter product/stock**

Element	Description
Scenario	Enter product/stock
Main Actor	Production, Administrator
Precondition	The user has access to the inventory module interface.
Description of the Scenario	Allows the user to register a new product or add stock units to an existing item in the inventory.
Main Flow	1. The actor selects the option to add stock or register a new product. 2. The system displays a form requesting item details (name, code, quantity, category). 3. The actor fills in the required fields and confirms. 4. The system validates the data, updates the database, and displays a success message.
Postcondition	The inventory stock levels are successfully updated.

Card 2: Edit product information

Element	Description
Scenario	Edit product information
Main Actor	Administrator
Preconditions	The product must already exist in the system database.
Description of the Scenario	Allows the Administrator to modify existing product details such as descriptions, names, or specifications to fix mistakes or keep data accurate.
Main Flow	1. The Administrator searches for the specific product. 2. The system displays the current details of the product in an editable form. 3. The Administrator modifies the necessary fields and clicks “Save changes”. 4. The system validates the inputs, updates the record, and confirms the action.
Postcondition	The product information is updated in the system database.

Card 3: Check availability

Element	Description
Scenario	Check availability
Main Actor	Administrator
Precondition	The system database is accessible.
Description of the Scenario	Allows the user to quickly query and view current stock status and quantities available for specific products.
Main Flow	1. The Administrator enters the product name or unique code in the search bar. 2. The system queries the database for matching records. 3. The system retrieves and displays the product name, code, and exact remaining stock units on screen.
Postcondition	The user successfully visualizes the availability of the queried items without modifying any data.

Database Entity-Relationship Diagram

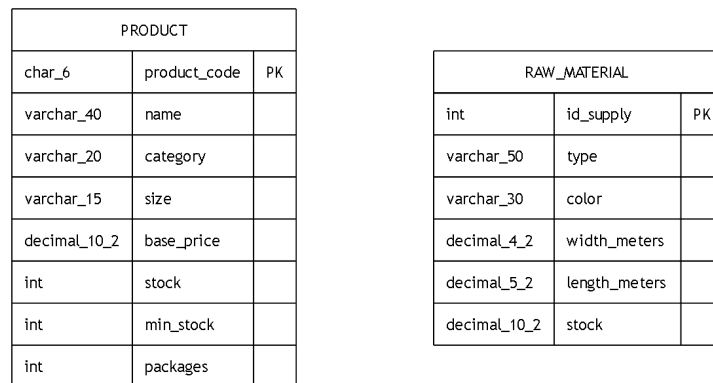


Figure 3: Entity-Relationship Diagram for Inventory Module

Main Entities

- **Products:** id_product (PK), product_code, name, category, size, base_price, stock_units, registration_date.
- **Inventory Logs / History:** id_log (PK), id_product (FK), user, date, action_type (Entry, Output, Manual Adjustment), quantity_changed, reason.

Interface Design (Mockups)

Main Inventory Dashboard Mockup



Figure 4: Main Inventory Menu - General Inventory view with Products, Materials, and Filters

Type of products Mockup

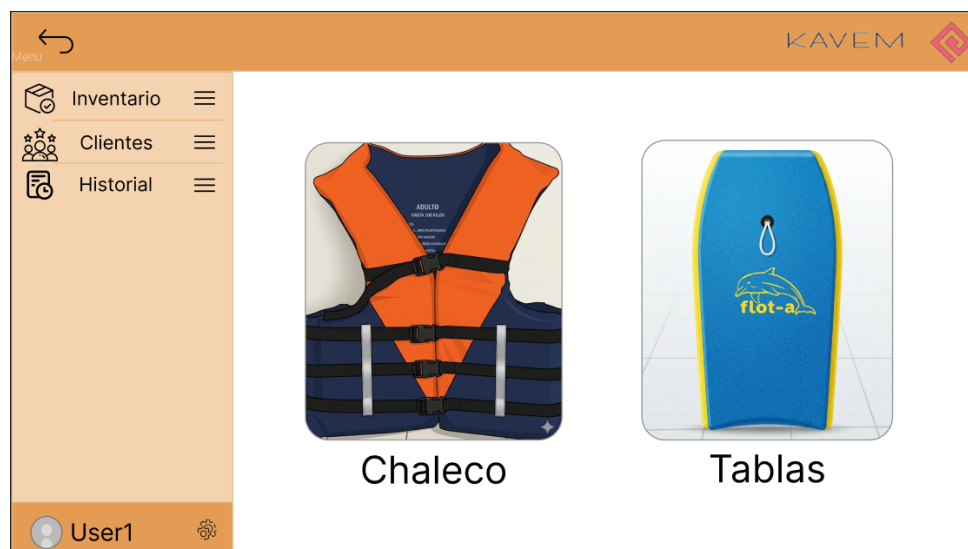


Figure 5: Type products Screen allows us to choose the type of product to consult

Product Details Mockup

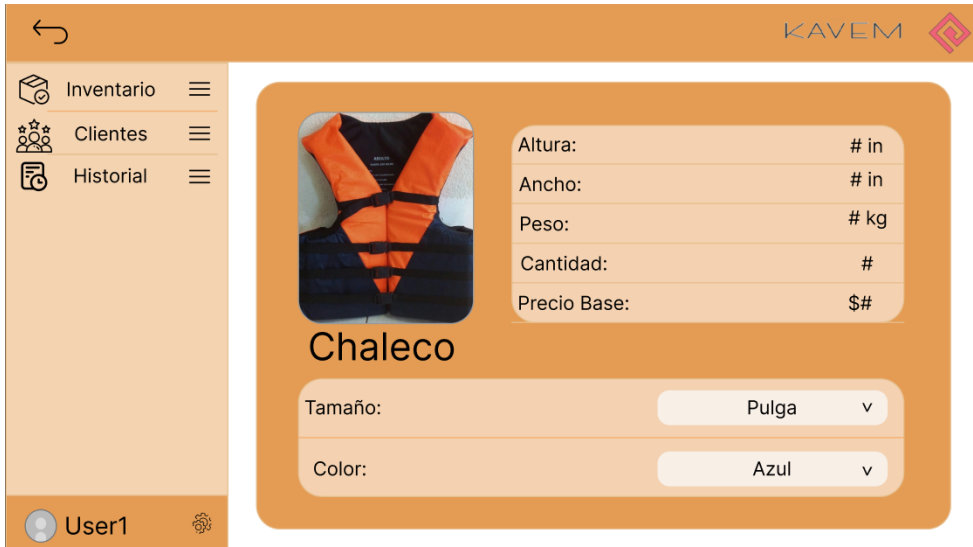


Figure 6: Product Details Screen displaying Name, Size, Category, Stock Level, and Price Settings

Raw materials Mockup



Figure 7: Raw materials Screen shows the available materials along with their characteristics

Navigation Design

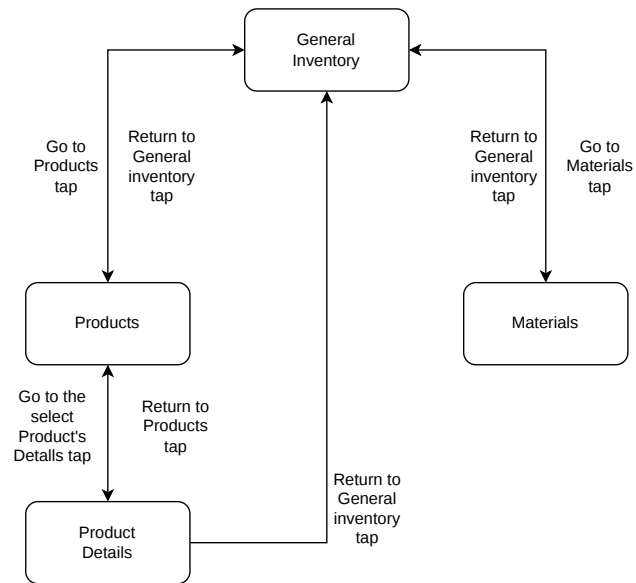


Figure 8: Navigation Flow: General Inventory Dashboard to Products and Materials Catalog

Navigation Structure

- **General Inventory Dashboard**

- Go to **Products** tab
 - * View General Products Catalog
 - * Search / Filter Products by attribute
 - * Go to **Product's Details** selection
 - View detailed tracking indicators
 - Return to Products list workflow
 - * Return to General Inventory
- Go to **Materials** tab
 - * View Raw Materials Inventory allocation
 - * Return to General Inventory dashboard

Test Cases

Sprint #4 Focus: Inventory Control operations (Enter Product, Handle Duplicates, Edit, and Check Availability)

Test Cases for Inventory Management

TC-01: Enter Product

Test Case ID	TC-01
Module	Inventory Module
Test Title	Enter Product - Stock Successfully
Priority	High
Preconditions	User has authenticated access to the inventory module interface
Test Data	Product details → producto_code -01, Name: Life Jacket, Quantity: 50
Execution Steps	1. Click on option to register product/add stock. 2. Input product details in requested input fields. 3. Confirm entry action.
Expected Results	Message: “Product successfully registered” and stock updates to 50.
Actual Results	As expected. Stock quantities reflected immediately.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 31, 2026

TC-02: Enter Product (Negative Test)

Test Case ID	TC-02
Module	Inventory Module
Test Title	Enter Product - Repeated Product ID
Priority	High
Preconditions	A product with identity registration code -01 must already exist
Test Data	Product ID → producto_code -01 (Repeated)
Execution Steps	1. Attempt to enter or create a new entry form. 2. Provide the repeated identifier string. 3. Submit and verify block constraint.

Expected Results	Message: “The ID is repeated”
Actual Results	As expected. Catalog blocks duplicate entry.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 31, 2026

TC-03: Edit Product

Test Case ID	TC-03
Module	Inventory Module
Test Title	Edit Product Information Successfully
Priority	High
Preconditions	Product with specified reference ID code must exist in DB
Test Data	Product modification → ID: producto_code -01, New Name: Premium Life Jacket
Execution Steps	1. Locate specific product inside system. 2. Edit name property and click “Save changes”.
Expected Results	Message: “Changes saved successfully” and the name updates in the system.
Actual Results	As expected. Data accurately refreshed in global catalogs.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 31, 2026

TC-04: Check Availability

Test Case ID	TC-04
Module	Inventory Module
Test Title	Check Stock Successfully
Priority	Medium
Preconditions	Target item must be part of active system records
Test Data	Search query → Code: producto_code-01
Execution Steps	1. Open look-up dashboard. 2. Fill search field bar input and validate criteria.

Expected Results	Display screen shows item state: Premium Life Jacket Stock: 50 units
Actual Results	As expected. Remaining numbers visible accurately.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 31, 2026

TC-05: Check Availability (Negative Test)

Test Case ID	TC-05
Module	Inventory Module
Test Title	Check Stock - Unregistered Product Criteria
Priority	Medium
Preconditions	None
Test Data	Search query → Code: FAKE-99
Execution Steps	1. Supply unknown key code or name to searching components. 2. Inspect fallback UI message.
Expected Results	Message: “No products found matching the criteria”
Actual Results	As expected. System avoids blank page error and renders warning label.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 31, 2026

Test Summary for Sprint #4

Test Case ID	Status
TC-01	Pass
TC-02	Pass
TC-03	Pass
TC-04	Pass
TC-05	Pass

Table 6: Test Cases completed in Sprint #4

Screenshots of the Program in Operation

Screenshot 1: Product Main

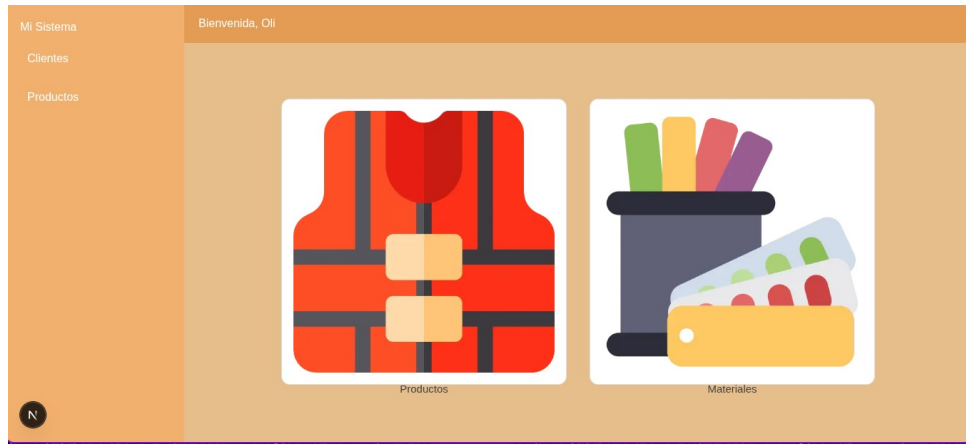


Figure 9: Product main shows the products that the company handles

Screenshot 2: Products Part

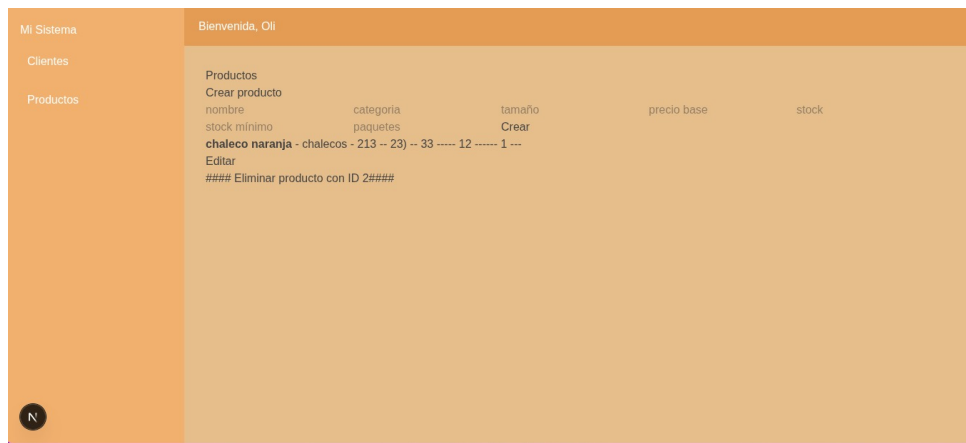


Figure 10: Products part allows us to add new products and edit and delete products

Screenshot 3: Materials

Mi Sistema

Clientes

Productos

Bienvenida, Oli

Materiales

Tipo de material

Color

Ancho en metros

Largo en metros

Stock

Crear

plastico - naranja - 12 ~ 12) ~ 2 ----

Editar

Eliminar material con ID 3####

N

Figure 11: Materials shows a description and quantity of the raw material

Sprint Management Using Jira

Sprint #3 Overview

Sprint Duration	May 28 – May 31, 2026
Total Tasks	8
Completed Tasks	8
Completion Percentage	100%

Jira Board Screenshot

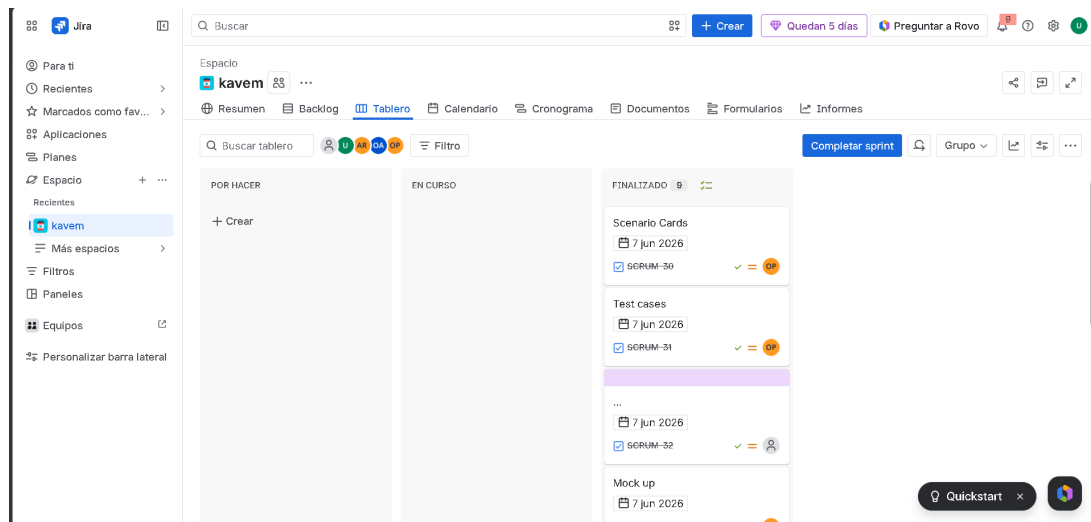


Figure 12: Jira Sprint Board for Sprint #4

Tasks Completed in Sprint #4

Issue ID	Task Name	Assigned To	Status
SCRUM-22	Update Functional Requirements (FR01-FR37)	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-23	Update Use Case Diagram	Omar Facio Palos	Finalizado
SCRUM-24	Update Scenario Cards	Omar Facio Palos	Finalizado
SCRUM-25	Update ER Diagram	Noé Ezequiel Estrada Ramírez	Finalizado
SCRUM-26	Refine Mockups (Main & Details)	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-27	Navigation Design Update	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-28	Execute Test Cases	Omar Facio Palos	Finalizado
SCRUM-29	Client Module Code Refinement	Olivia Lizbeth Chairez Alba	Finalizado

Table 7: Tasks completed during Sprint #4

Task Assignments Summary

- **Olivia Lizbeth Chairez Alba (UP240594)** - Client Module Code Refinement (SCRUM-29)
- **Noé Ezequiel Estrada Ramírez (UP240770)** - Update ER Diagram (SCRUM-25)
- **Omar Facio Palos (UP240470)** - Update Use Case Diagram (SCRUM-23), Update Scenario Cards (SCRUM-24), Test Cases Execution (SCRUM-28)
- **Alan Ricardo Jauregui De La Riva (UP240960)** - Update Requirements (SCRUM-22), refine Mockups (SCRUM-26), Navigation Update (SCRUM-27), Test Cases Execution (SCRUM-28)

Burndown Chart

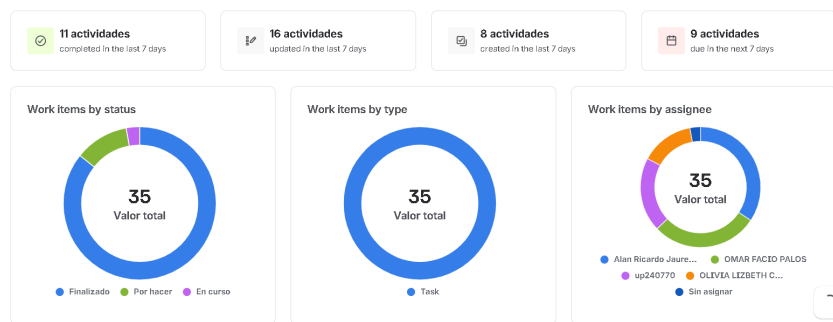


Figure 13: Sprint #4 Burndown Chart from Jira